VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

SCHOOL OF ELECTRONICS ENGINEERING

Course : Antenna and Microwave Engineering Lab (BECE305L)
Lab Slot : L23+L24
Faculty Name : YOGESH KUMAR CHOUKIKER

Task/ Experiment No. : 2	**Date : 5/09/2025**
Name of the Student : HARI KRUTHIK SK	**Regd. No. : 23BEC0358**

WILKILSON POWER DIVIDER [EQUAL] and [UNEQUAL]:

CASE 1)

ε = 2.2
H = 1.58 mm
F = 3.5 GHz
t = 0.035 mm
Z0 = 50 ohms

CASE 2)

K^2 = 0.5
ε = 4.4
H = 1.6 mm
F = 3GHz
t = 0.035 mm
Z0 = 50 ohms

AIM :

- To design and simulate a Wilkinson power divider for equal and unequal power divisions.
- To determine the scattering parameters of Wilkinson power divider.

SOFTWARE USED :

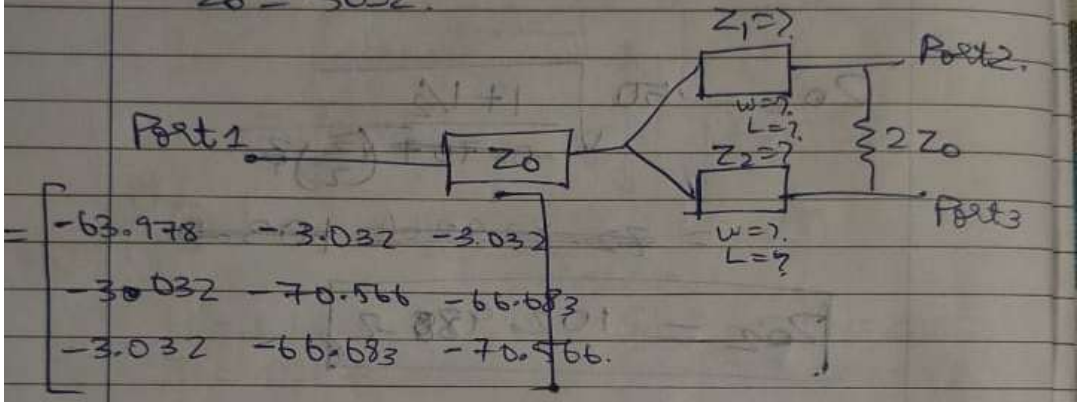
AWR Design Environment (Microwave Office) by **Cadence (formerly by NI/AWR)**

1/7/25

Date / /

Wilkinson Power Divider (equal)

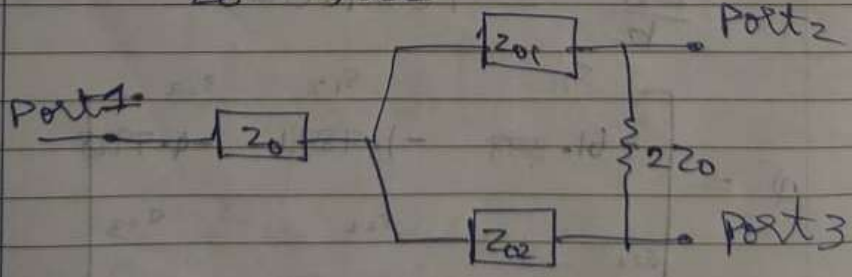
$\varepsilon = 2.2$
$H = 1.58mm$
$F = 3.5\ GHz$
$t = 0.035mm$
$Z_0 = 50\Omega$.

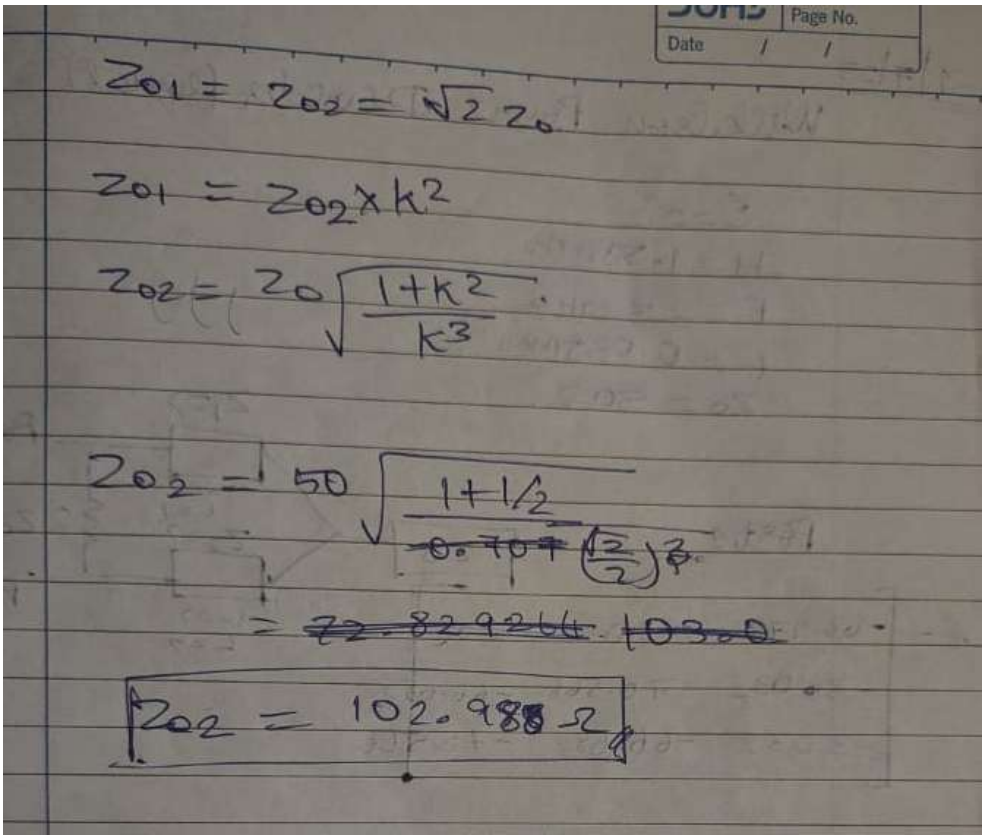


$$= \begin{bmatrix} -63.978 & -3.032 & -3.032 \\ -3.032 & -70.566 & -66.683 \\ -3.032 & -66.683 & -70.566 \end{bmatrix}$$

Unequal case:

$K^2 = 1/2$
$\varepsilon_r = 4.4$
$H = 1.6\ mm$
$F = 3\ GHz$
$t = 0.035\ mm$
$Z_0 = 50\Omega$.





$$Z_{01} = Z_{02} = \sqrt{2}\, Z_0$$

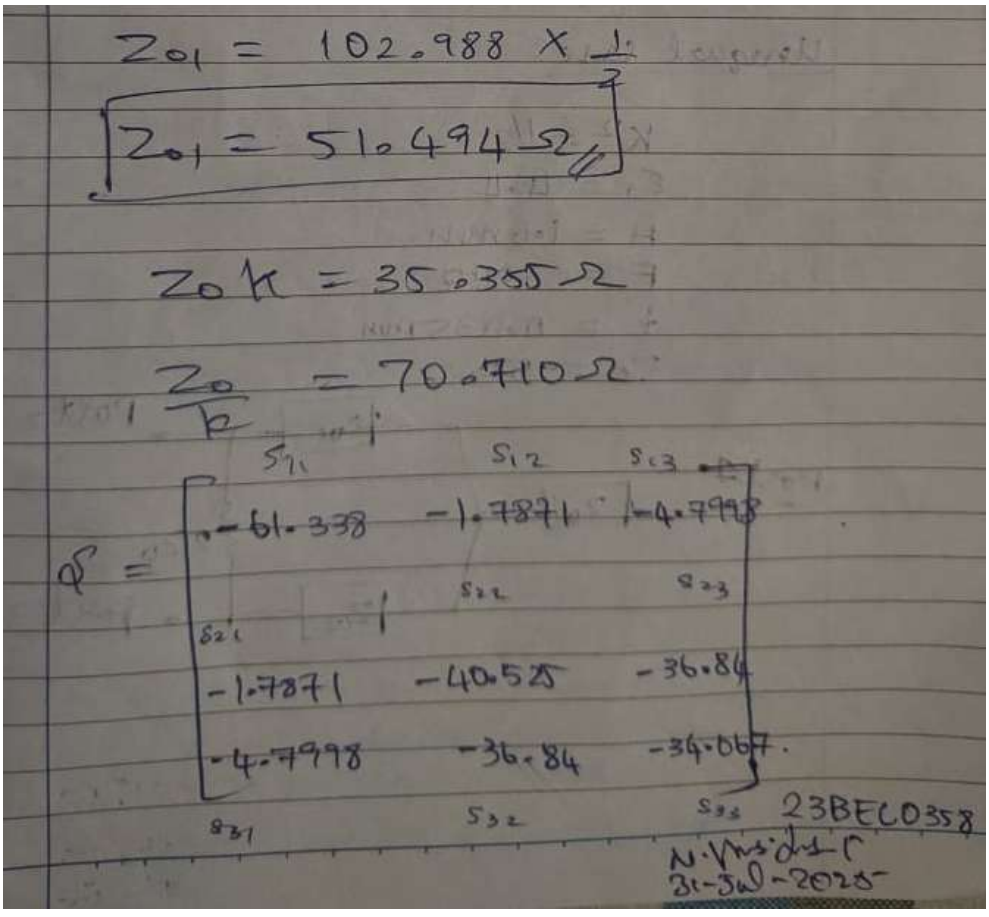
$$Z_{01} = Z_{02} \times k^2$$

$$Z_{02} = Z_0 \sqrt{\frac{1+k^2}{k^3}}$$

$$Z_{02} = 50 \sqrt{\frac{1+1/2}{0.707 \left(\frac{1}{2}\right)^3}}$$

$$= 72.82926 \times 103.0$$

$$\boxed{Z_{02} = 102.988\ \Omega}$$



$$Z_{01} = 102.988 \times \frac{1}{2}$$

$$\boxed{Z_{01} = 51.494\ \Omega}$$

$$Z_0 k = 35.355\ \Omega$$

$$\frac{Z_0}{k} = 70.710\ \Omega$$

$$S = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ -61.338 & -1.7871 & -4.7998 \\ S_{21} & S_{22} & S_{23} \\ -1.7871 & -40.525 & -36.84 \\ -4.7998 & -36.84 & -34.067 \\ S_{31} & S_{32} & S_{33} \end{bmatrix}$$

23BEC0358
N. Vrushik
31-3-2025

Expt. No. : 2

Date : 31/7/2025

Wilkinson Power Divider

Objectives:

1. To design and simulation of a Wilknson power divider for equal and unequal power divisions.
2. To determine the scattering parameters of Wilkinson power divider.

Equipment required: AWR Microwave Office software

Specifications :

	Equal Case	Unequal Case
Characteristic impedance	$Z_0 = 50\Omega$	$Z_0 = 50\Omega$
Operating frequency	$f = 3.5$ GHz	$f = 3$GHz
Substrate thickness	$H = 1.58$ mm	$H = 1.6$mm
Metal thickness	$T = 0.035$ mm	$T = 0.035$mm
Dielectric constant	$\varepsilon_r = 2.2$	$\varepsilon_r = 4.4$
Loss tangent	$L = 0.02$	$K^2 = 1/2$
		$L = 0.02$

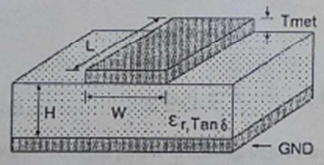


Figure (1) Microstrip Line

Theory: The Wilkinson power divider is generally designed using microstrip lines as shown in figure 2 and can be made with any number of ports with equal or unequal power divisions. Wilkinson power divider has many advantages over other power dividers and has the following properties

1. Matched at all ports.
2. Large isolation between output ports
3. Reciprocal
4. Loss less when output ports are matched

$$[S] = \begin{pmatrix} 0 & \frac{-j}{\sqrt{2}} & \frac{-j}{\sqrt{2}} \\ \frac{-j}{\sqrt{2}} & 0 & 0 \\ \frac{-j}{\sqrt{2}} & 0 & 0 \end{pmatrix}$$

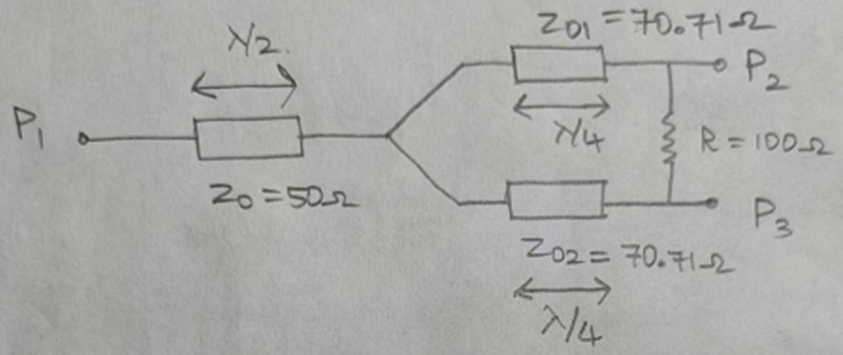
The S-matrix of a 3-port Wilkinson power divider is given by

Equal Case:

$Z_0 = 50\Omega$

$Z_{01} = Z_{02} = \sqrt{2} \times Z_0 = 70.71\Omega$.

$R = 2Z_0 = 100\Omega$.



*) $Z = 50\Omega$

$L = 31.1045$ mm

$W = 4.8321$ mm.

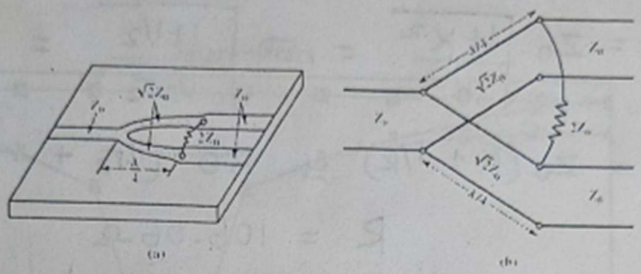
*) $Z = 70.71\Omega$

$L = 15.8839$mm

$W = 2.074029$mm

$$S = \begin{bmatrix} 64.003 & 3.032 & 3.032 \\ 3.032 & 70.529 & 66.614 \\ 3.032 & 66.614 & 70.529 \end{bmatrix}$$

Figure (2)



Design Equations:

$$\text{Power division Ratio} = K^2 = \frac{P_3}{P_2}$$

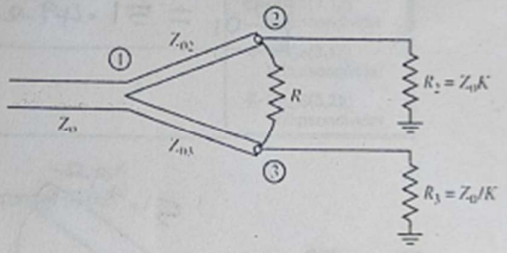
$$Z_{03} = Z_0 \sqrt{\frac{1+K^2}{K^3}}$$

$$Z_{02} = Z_{03}\ K^2 = Z_0 \sqrt{K(1+k^2)}$$

$$R = Z_0\ (K + \frac{1}{K}\ )$$

$$R_2 = Z_0 K$$

$$R_3 = Z_0 / K$$



Sample Observations: For equal power division,sample results of a Wilkinson power divider shown below

Z_0= 50Ω, f=3GHz, H=1.6mm, T=0.036 mm, ε_r=4.4, L=0.001

Table : S-Parameters in dB

Frequency (GHz)	DB(S(1,1)) Wilkinson divider	DB(S(2,1)) Wilkinson divider	DB(S(3,1)) Wilkinson divider	DB(S(3,2)) Wilkinson divider
1	-10.711	-3.4135	-3.4135	-8.0407
1.5	-12.359	-3.2914	-3.2914	-11.02
2	-15.24	-3.1664	-3.1664	-14.659
2.5	-20.842	-3.0737	-3.0737	-20.645
3	-61.983	-3.0112	-3.0112	-64.769
3.5	-20.668	-3.0824	-3.0824	-20.573
4	-15.007	-3.1686	-3.1686	-14.546
4.5	-12.131	-3.3284	-3.3284	-10.901
5	-10.559	-3.4588	-3.4588	-7.896
5.5	-9.7996	-3.5443	-3.5443	-4.9625
6	-9.6207	-3.5698	-3.5698	-3.6236

Unequal case:

$Z_0 = 50\Omega$ $\quad K^2 = 1/2$

$K = 1/\sqrt{2}$

$$Z_{02} = Z_0\sqrt{\frac{1+K^2}{K^3}} = 50\sqrt{\frac{1+1/2}{1/2\sqrt{2}}} = 102.98\Omega$$

$$R = Z_0(K + 1/K) = 50\left(1/\sqrt{2} + 1/1/\sqrt{2}\right)$$

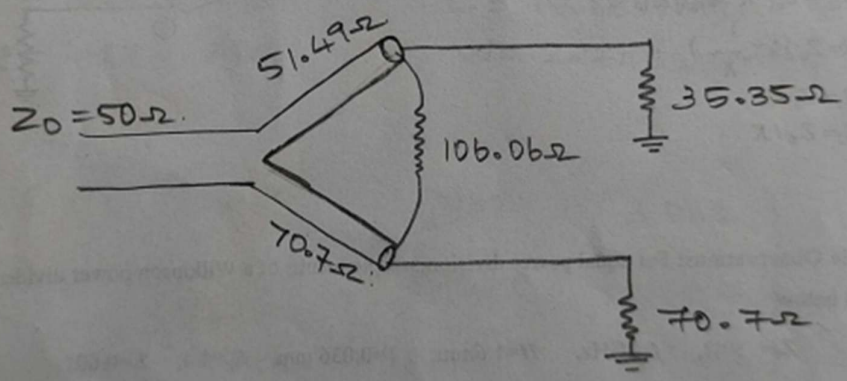
$$R = 106.06\Omega$$

$$Z_{01} = Z_{02}K^2 = 102.98 \times \frac{1}{2}$$

$$\boxed{Z_{01} = 51.49\Omega}$$

$Z_0K = 35.35\Omega$

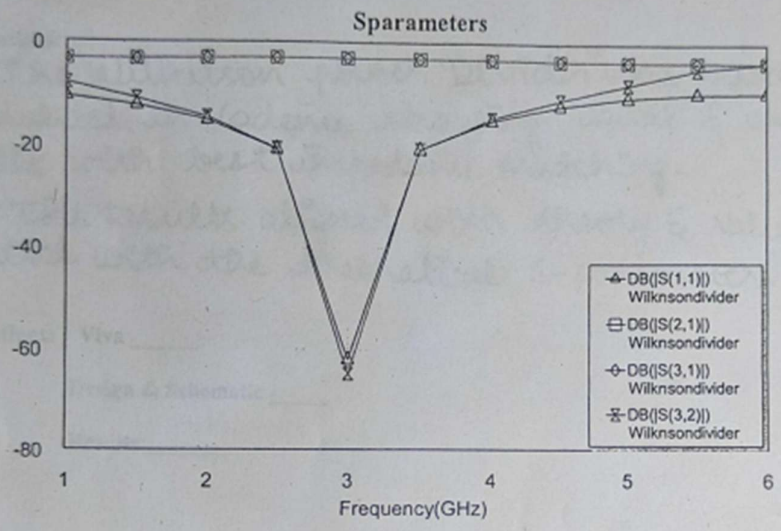
$\frac{Z_0}{K} = 70.7\Omega$



$$S = \begin{bmatrix} 61.338 & 1.7871 & 4.7998 \\ 1.7891 & 40.525 & 36.84 \\ 4.7998 & 36.84 & 34.067 \end{bmatrix}$$

Model graph:

S-Parameter Plot



Practical Observations:

Equal case

Frequency ~~Mhz~~/Ghz	S_{11}	S_{21}	S_{31}	S_{32}
3.47	-47.651	-3.032	-3.032	-45.984
3.48	-51.84	-3.0319	-3.0319	-49.28
3.49	-60.2	-3.0319	-3.0319	-54.594
f → 3.5	-64.003	-3.032	-3.032	-66.614
3.51	-53.124	-3.032	-3.032	-56.903
3.52	-48.425	-3.0321	-3.0321	-50.487
3.53	-45.292	-3.0322	-3.0322	-46.796

Unequal case

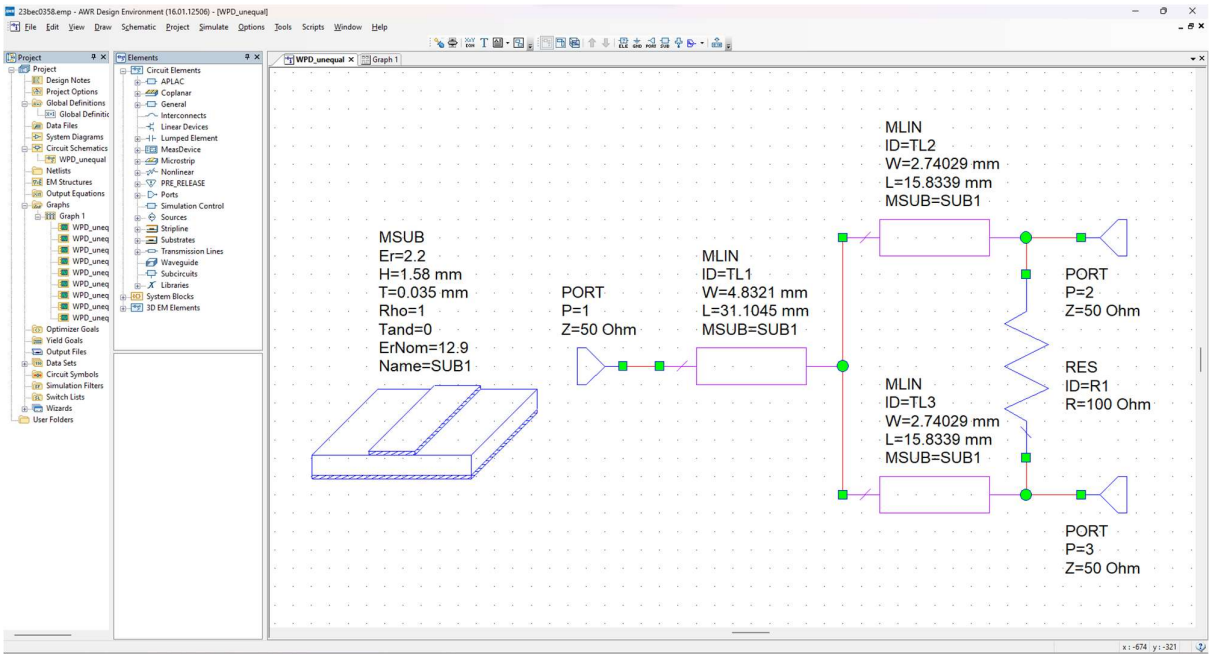
Frequency GHz	S11	S21	S31	S32
2.94	-45.595	-1.787	-4.7999	-36.202
2.98	-49.819	-1.787	-4.7998	-36.532
2.99	-58.292	-1.787	-4.7998	-36.75
f → 3	-61.338	-1.7871	-4.7998	-36.86
3.01	-50.873	-1.7871	-4.7998	-36.792
3.02	-46.233	-1.7873	-4.7999	-36.611
3.03	-43.233	-1.7874	-4.8	-36.312

Conclusions:

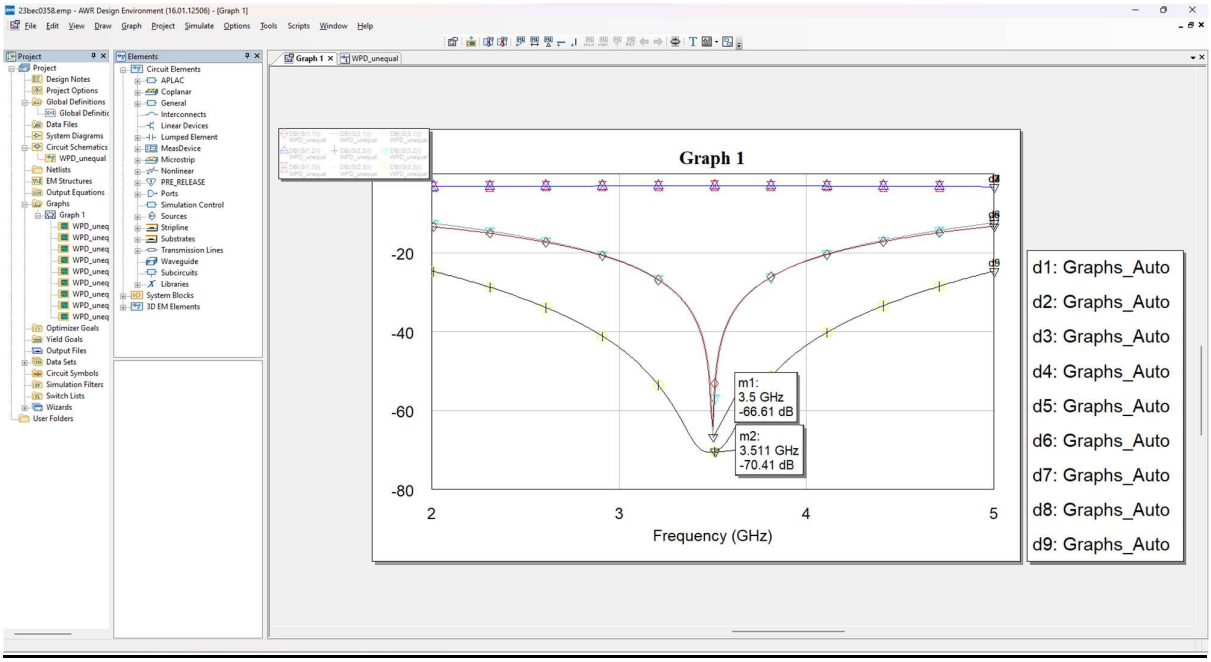
1) The Wilkilson power Divider was successfully simulated in Cadence showing equal & unequal power split with best impedance matching.

2) The Results aligned with theory & values are united with the theoratical S-parameter.

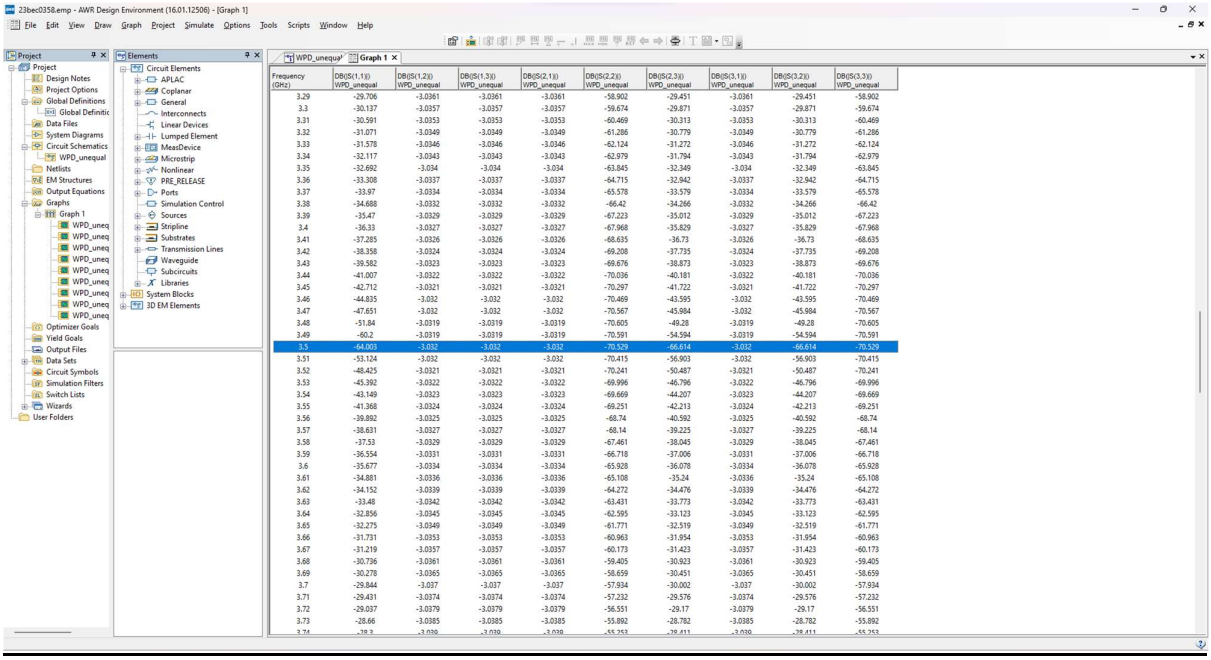
SCHEMATIC DIAGRAM : [EQUAL]



GRAPH :

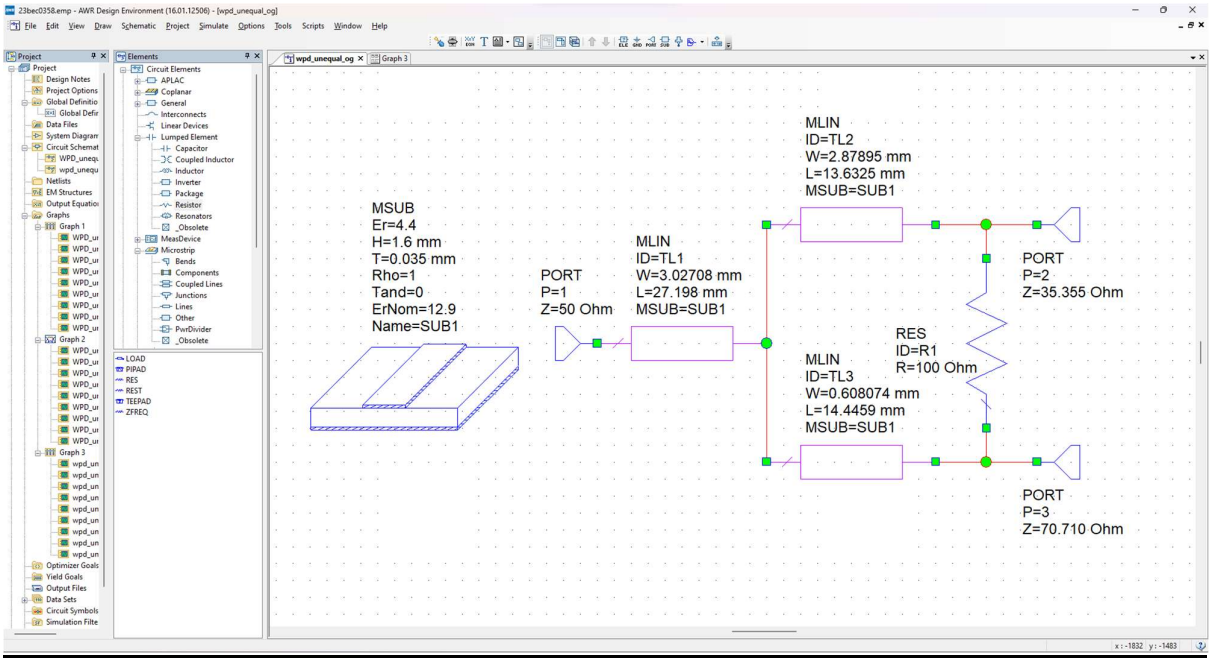


TABULAR COLUMN :

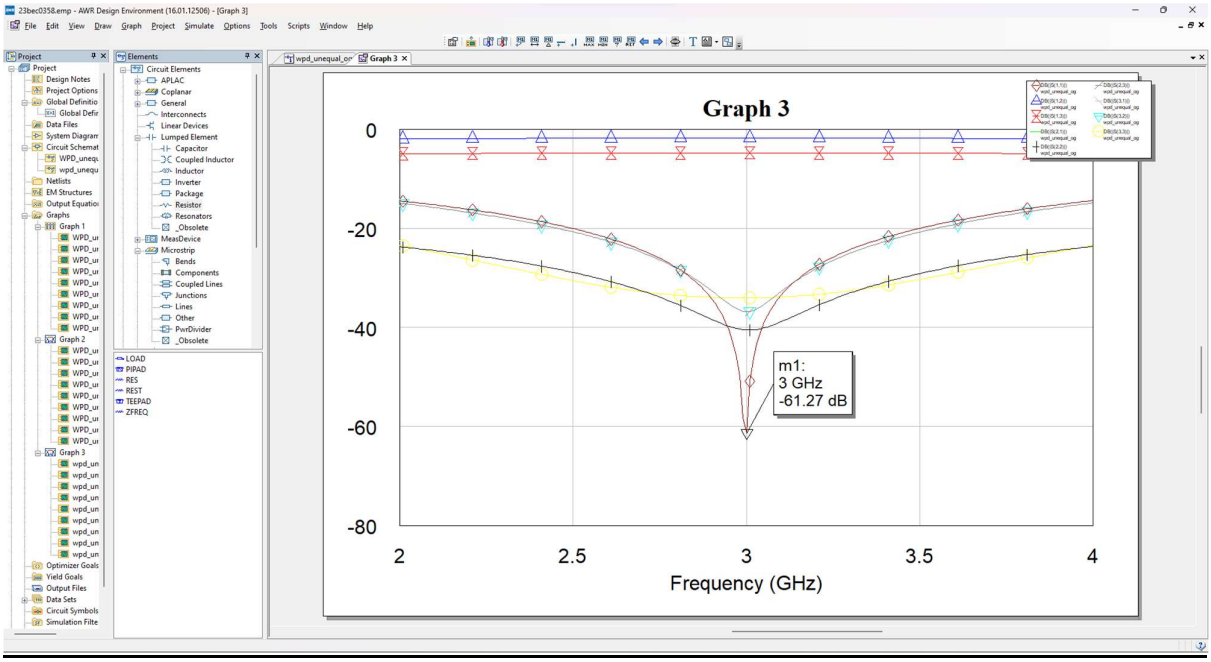


Frequency (GHz)	DB(S(1,1)) WPD_unequal	DB(S(1,2)) WPD_unequal	DB(S(1,3)) WPD_unequal	DB(S(2,1)) WPD_unequal	DB(S(2,2)) WPD_unequal	DB(S(2,3)) WPD_unequal	DB(S(3,1)) WPD_unequal	DB(S(3,2)) WPD_unequal	DB(S(3,3)) WPD_unequal
3.29	-29.706	-3.0361	-3.0361	-3.0361	-58.902	-29.451	-3.0361	-29.451	-58.902
3.3	-30.137	-3.0357	-3.0357	-3.0357	-59.674	-29.871	-3.0357	-29.871	-59.674
3.31	-30.591	-3.0353	-3.0353	-3.0353	-60.469	-30.313	-3.0353	-30.313	-60.469
3.32	-31.071	-3.0349	-3.0349	-3.0349	-61.286	-30.779	-3.0349	-30.779	-61.286
3.33	-31.578	-3.0346	-3.0346	-3.0346	-62.124	-31.272	-3.0346	-31.272	-62.124
3.34	-32.117	-3.0343	-3.0343	-3.0343	-62.979	-31.794	-3.0343	-31.794	-62.979
3.35	-32.692	-3.034	-3.034	-3.034	-63.845	-32.349	-3.034	-32.349	-63.845
3.36	-33.308	-3.0337	-3.0337	-3.0337	-64.715	-32.942	-3.0337	-32.942	-64.715
3.37	-33.97	-3.0334	-3.0334	-3.0334	-65.578	-33.579	-3.0334	-33.579	-65.578
3.38	-34.688	-3.0332	-3.0332	-3.0332	-66.42	-34.266	-3.0332	-34.266	-66.42
3.39	-35.47	-3.0329	-3.0329	-3.0329	-67.223	-35.012	-3.0329	-35.012	-67.223
3.4	-36.33	-3.0327	-3.0327	-3.0327	-67.968	-35.829	-3.0327	-35.829	-67.968
3.41	-37.285	-3.0326	-3.0326	-3.0326	-68.635	-36.73	-3.0326	-36.73	-68.635
3.42	-38.358	-3.0324	-3.0324	-3.0324	-69.208	-37.735	-3.0324	-37.735	-69.208
3.43	-39.582	-3.0323	-3.0323	-3.0323	-69.676	-38.873	-3.0323	-38.873	-69.676
3.44	-41.007	-3.0322	-3.0322	-3.0322	-70.036	-40.181	-3.0322	-40.181	-70.036
3.45	-42.712	-3.0321	-3.0321	-3.0321	-70.297	-41.722	-3.0321	-41.722	-70.297
3.46	-44.835	-3.032	-3.032	-3.032	-70.469	-43.595	-3.032	-43.595	-70.469
3.47	-47.651	-3.032	-3.032	-3.032	-70.567	-45.984	-3.032	-45.984	-70.567
3.48	-51.84	-3.0319	-3.0319	-3.0319	-70.605	-49.28	-3.0319	-49.28	-70.605
3.49	-60.2	-3.0319	-3.0319	-3.0319	-70.591	-54.594	-3.0319	-54.594	-70.591
3.5	-64.003	-3.032	-3.032	-3.032	-70.529	-66.614	-3.032	-66.614	-70.529
3.51	-53.124	-3.032	-3.032	-3.032	-70.415	-56.903	-3.032	-56.903	-70.415
3.52	-48.425	-3.0321	-3.0321	-3.0321	-70.241	-50.487	-3.0321	-50.487	-70.241
3.53	-45.392	-3.0322	-3.0322	-3.0322	-69.996	-46.796	-3.0322	-46.796	-69.996
3.54	-43.149	-3.0323	-3.0323	-3.0323	-69.669	-44.207	-3.0323	-44.207	-69.669
3.55	-41.368	-3.0324	-3.0324	-3.0324	-69.251	-42.213	-3.0324	-42.213	-69.251
3.56	-39.892	-3.0325	-3.0325	-3.0325	-68.74	-40.592	-3.0325	-40.592	-68.74
3.57	-38.631	-3.0327	-3.0327	-3.0327	-68.14	-39.225	-3.0327	-39.225	-68.14
3.58	-37.53	-3.0329	-3.0329	-3.0329	-67.461	-38.045	-3.0329	-38.045	-67.461
3.59	-36.554	-3.0331	-3.0331	-3.0331	-66.718	-37.006	-3.0331	-37.006	-66.718
3.6	-35.677	-3.0334	-3.0334	-3.0334	-65.928	-36.078	-3.0334	-36.078	-65.928
3.61	-34.881	-3.0336	-3.0336	-3.0336	-65.108	-35.24	-3.0336	-35.24	-65.108
3.62	-34.152	-3.0339	-3.0339	-3.0339	-64.272	-34.476	-3.0339	-34.476	-64.272
3.63	-33.48	-3.0342	-3.0342	-3.0342	-63.431	-33.773	-3.0342	-33.773	-63.431
3.64	-32.856	-3.0345	-3.0345	-3.0345	-62.595	-33.123	-3.0345	-33.123	-62.595
3.65	-32.275	-3.0349	-3.0349	-3.0349	-61.771	-32.519	-3.0349	-32.519	-61.771
3.66	-31.731	-3.0353	-3.0353	-3.0353	-60.963	-31.954	-3.0353	-31.954	-60.963
3.67	-31.219	-3.0357	-3.0357	-3.0357	-60.173	-31.423	-3.0357	-31.423	-60.173
3.68	-30.736	-3.0361	-3.0361	-3.0361	-59.405	-30.923	-3.0361	-30.923	-59.405
3.69	-30.278	-3.0365	-3.0365	-3.0365	-58.659	-30.451	-3.0365	-30.451	-58.659
3.7	-29.844	-3.037	-3.037	-3.037	-57.934	-30.002	-3.037	-30.002	-57.934
3.71	-29.431	-3.0374	-3.0374	-3.0374	-57.232	-29.576	-3.0374	-29.576	-57.232
3.72	-29.037	-3.0379	-3.0379	-3.0379	-56.551	-29.17	-3.0379	-29.17	-56.551
3.73	-28.66	-3.0385	-3.0385	-3.0385	-55.892	-28.782	-3.0385	-28.782	-55.892

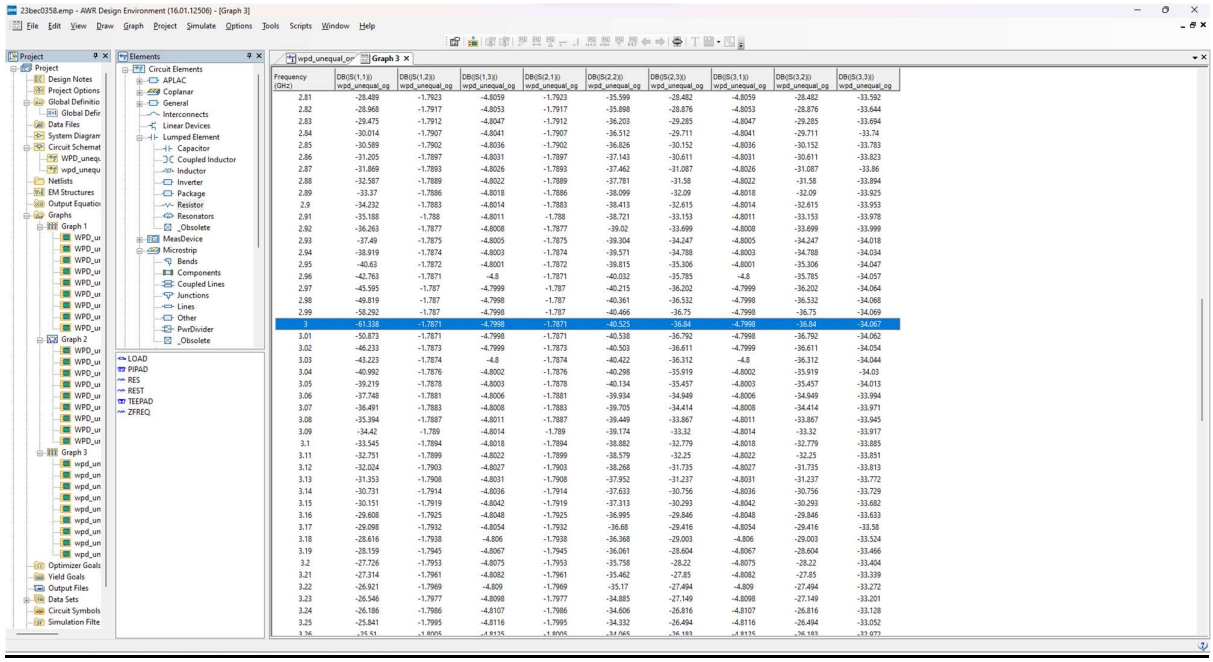
SCHEMATIC DIAGRAM : [UNEQUAL]



GRAPH :



TABULAR COLUMN :



Frequency (GHz)	DB(S(1,1)) wpd_unequal_og	DB(S(1,2)) wpd_unequal_og	DB(S(1,3)) wpd_unequal_og	DB(S(2,1)) wpd_unequal_og	DB(S(2,2)) wpd_unequal_og	DB(S(2,3)) wpd_unequal_og	DB(S(3,1)) wpd_unequal_og	DB(S(3,2)) wpd_unequal_og	DB(S(3,3)) wpd_unequal_og
2.81	-28.489	-1.7923	-4.8059	-1.7923	-35.599	-28.482	-4.8059	-28.482	-33.592
2.82	-28.968	-1.7917	-4.8053	-1.7917	-35.898	-28.876	-4.8053	-28.876	-33.644
2.83	-29.475	-1.7912	-4.8047	-1.7912	-36.203	-29.285	-4.8047	-29.285	-33.694
2.84	-30.014	-1.7907	-4.8041	-1.7907	-36.512	-29.711	-4.8041	-29.711	-33.74
2.85	-30.589	-1.7902	-4.8036	-1.7902	-36.826	-30.152	-4.8036	-30.152	-33.783
2.86	-31.205	-1.7897	-4.8031	-1.7897	-37.143	-30.611	-4.8031	-30.611	-33.823
2.87	-31.869	-1.7893	-4.8026	-1.7893	-37.462	-31.087	-4.8026	-31.087	-33.86
2.88	-32.587	-1.7889	-4.8022	-1.7889	-37.781	-31.58	-4.8022	-31.58	-33.894
2.89	-33.37	-1.7886	-4.8018	-1.7886	-38.099	-32.09	-4.8018	-32.09	-33.925
2.9	-34.232	-1.7883	-4.8014	-1.7883	-38.413	-32.615	-4.8014	-32.615	-33.953
2.91	-35.188	-1.788	-4.8011	-1.788	-38.721	-33.153	-4.8011	-33.153	-33.978
2.92	-36.263	-1.7877	-4.8008	-1.7877	-39.02	-33.699	-4.8008	-33.699	-33.999
2.93	-37.49	-1.7875	-4.8005	-1.7875	-39.304	-34.247	-4.8005	-34.247	-34.018
2.94	-38.919	-1.7874	-4.8003	-1.7874	-39.571	-34.788	-4.8003	-34.788	-34.034
2.95	-40.63	-1.7872	-4.8001	-1.7872	-39.815	-35.306	-4.8001	-35.306	-34.047
2.96	-42.763	-1.7871	-4.8	-1.7871	-40.032	-35.785	-4.8	-35.785	-34.057
2.97	-45.595	-1.787	-4.7999	-1.787	-40.215	-36.202	-4.7999	-36.202	-34.064
2.98	-49.819	-1.787	-4.7998	-1.787	-40.361	-36.532	-4.7998	-36.532	-34.068
2.99	-58.292	-1.787	-4.7998	-1.787	-40.466	-36.75	-4.7998	-36.75	-34.069
3	-61.338	-1.7871	-4.7998	-1.7871	-40.525	-36.84	-4.7998	-36.84	-34.067
3.01	-50.873	-1.7871	-4.7998	-1.7871	-40.538	-36.792	-4.7998	-36.792	-34.062
3.02	-46.233	-1.7873	-4.7999	-1.7873	-40.503	-36.611	-4.7999	-36.611	-34.054
3.03	-43.223	-1.7874	-4.8	-1.7874	-40.422	-36.312	-4.8	-36.312	-34.044
3.04	-40.992	-1.7876	-4.8002	-1.7876	-40.298	-35.919	-4.8002	-35.919	-34.03
3.05	-39.219	-1.7878	-4.8003	-1.7878	-40.134	-35.457	-4.8003	-35.457	-34.013
3.06	-37.748	-1.7881	-4.8006	-1.7881	-39.934	-34.949	-4.8006	-34.949	-33.994
3.07	-36.491	-1.7883	-4.8008	-1.7883	-39.705	-34.414	-4.8008	-34.414	-33.971
3.08	-35.394	-1.7887	-4.8011	-1.7887	-39.449	-33.867	-4.8011	-33.867	-33.945
3.09	-34.42	-1.789	-4.8014	-1.789	-39.174	-33.32	-4.8014	-33.32	-33.917
3.1	-33.545	-1.7894	-4.8018	-1.7894	-38.882	-32.779	-4.8018	-32.779	-33.885
3.11	-32.751	-1.7899	-4.8022	-1.7899	-38.579	-32.25	-4.8022	-32.25	-33.851
3.12	-32.024	-1.7903	-4.8027	-1.7903	-38.268	-31.735	-4.8027	-31.735	-33.813
3.13	-31.353	-1.7908	-4.8031	-1.7908	-37.952	-31.237	-4.8031	-31.237	-33.772
3.14	-30.731	-1.7914	-4.8036	-1.7914	-37.633	-30.756	-4.8036	-30.756	-33.729
3.15	-30.151	-1.7919	-4.8042	-1.7919	-37.313	-30.293	-4.8042	-30.293	-33.682
3.16	-29.608	-1.7925	-4.8048	-1.7925	-36.995	-29.846	-4.8048	-29.846	-33.633
3.17	-29.098	-1.7932	-4.8054	-1.7932	-36.68	-29.416	-4.8054	-29.416	-33.58
3.18	-28.616	-1.7938	-4.806	-1.7938	-36.368	-29.003	-4.806	-29.003	-33.524
3.19	-28.159	-1.7945	-4.8067	-1.7945	-36.061	-28.604	-4.8067	-28.604	-33.466
3.2	-27.726	-1.7953	-4.8075	-1.7953	-35.758	-28.22	-4.8075	-28.22	-33.404
3.21	-27.314	-1.7961	-4.8082	-1.7961	-35.462	-27.85	-4.8082	-27.85	-33.339
3.22	-26.921	-1.7969	-4.809	-1.7969	-35.17	-27.494	-4.809	-27.494	-33.272
3.23	-26.546	-1.7977	-4.8098	-1.7977	-34.885	-27.149	-4.8098	-27.149	-33.201
3.24	-26.186	-1.7986	-4.8107	-1.7986	-34.606	-26.816	-4.8107	-26.816	-33.128
3.25	-25.841	-1.7995	-4.8116	-1.7995	-34.332	-26.494	-4.8116	-26.494	-33.052
3.26	-25.51	-1.8005	-4.8125	-1.8005	-34.065	-26.183	-4.8125	-26.183	-32.972